5.1 Real Numbers

Definition 5.1.1 (Ordered Ring / Field)
*An **ordered ring** is a ring $(A, +, \cdot)$ such that the additive group $(A, +)$ is an ordered abelian group with ordering $\leq$ which also satisfies*

$$x, y \geq 0 \implies x \cdot y \geq 0$$

*An **ordered field** is a field with the structure of an ordered ring.*

Proposition 5.1.2 (Extension to Field of Fractions)
Let A be an ordered ring with no zero-divisors. The $K = \text{Frac}(A)$ is an ordered ring with the order given by

$$\frac{x}{y} \leq \frac{w}{z} \iff xz \leq wy$$

Example 5.1.3
$\mathbb{Z}$ is an ordered ring and $\mathbb{Q} := \text{Frac}(\mathbb{Z})$ is an ordered field.

Proposition 5.1.4 (Trichotomy Law)
Let G be an ordered abelian group. Then G is the disjoint union of $\{0\}$, G^+ and G^-.

More generally precisely one of $x = y$, $x < y$ and $x > y$ holds.

Lemma 5.1.5 (Elementary Properties)
Let G be an ordered abelian group. Then

a) $x \in G^+ \iff -x \in G^-$
b) $x \in G^+$ and $x \leq y \implies y \in G^+$
c) $x \in G^-$ and $y \leq x \implies x \in G^-$
d) $x, y \in G^+ \implies x + y \in G^+$
e) $x, y \in G^- \implies x + y \in G^-$
f) $x, y \in G^+ \implies xy \in G^+$

Proof.
a) $x \in G^+ \implies 0 \leq x \implies -x \leq 0$. Furthermore $x \neq 0 \implies -x \neq 0$ by assumption so $-x \in G^-$.
b) By transitivity $0 \leq y$. Suppose $y = 0$ then $0 \leq x \leq 0$ so $x = 0$ a contradiction.
c) Similarly
d) Follows from b) since $0 \leq x \implies 0 \leq y \leq x + y$.
e) Follows from c) since $x \leq 0 \implies x + y \leq y \leq 0$.
f) If $xy = 0 \implies x = 0$ or $y = 0$ which is a contradiction. Therefore $xy > 0$.

■

Proposition 5.1.6 (Ordered Rings are Torsion Free)
Let A be a non-zero ordered ring. Then the canonical map $\mathbb{Z} \to A$ is injective, in other words A has characteristic 0.

In particular every ordered field K contains $\mathbb{Q}$ as a subring.

Proof. Follows immediately by induction, since if $n.1 = 0$ then $(n-1) > n \implies 1 < 0 \implies -1 > 0 \implies 1 = -1 * -1 < 0$ a contradiction. ■

Proposition 5.1.7 (Archimedean Field)
Let K be an ordered field. Then the following are equivalent

a) $\mathbb{Q}$ is dense in K i.e. for all $x, y \in K$ there exists $z \in \mathbb{Q}$ such that $x < z < y$
b) For all $x, y \in K$ there exists $n \in \mathbb{N}$ such that $ny > x$

*In this case we say K is **Archimedean**.*

In particular for every $\varepsilon \in K^+$ there exists $n \in \mathbb{N}^+$ such that $\frac{1}{n} < \varepsilon$.

Proof. Suppose the second property is satisfied then choose $n > \frac{1}{y-x}$. By definition $0 < \frac{1}{n} < y - x$. Let

$$S = \left\{ m \in \mathbb{Z} \;\middle|\; \frac{m}{n} \leq x \right\}$$

By the Archimedean property (applied to $\frac{1}{n}$ and $-x$) S is non-empty. By the well-ordering principle it has a maximal element. By maximality $\frac{m+1}{n} > x$. Furthermore

© David Rufino CC BY-NC-ND 4.0

$$x < \frac{m+1}{n} \leq x + \frac{1}{n} < x + (y - x) = y$$

and therefore $z \coloneqq \frac{m+1}{n}$ satisfies the required property.

Conversely suppose K satisfies the first property and assume wlog that $x, y > 0$. Choose $\frac{m}{n}$ such that

$$0 < \frac{m}{n} < \frac{y}{x}$$

which implies $ny > mx \geq x$. ■

Definition 5.1.8 (Absolute Value)
*Let G be an **ordered abelian group**. Define the absolute value as*

$$|x| \coloneqq \begin{cases} x & x \in G^+ \\ -x & x \in G^- \\ 0 & x = 0 \end{cases}$$

Then this satisfies the following properties

a) $|x| \geq 0$
b) $|-x| = |x|$
c) $|x| = 0 \iff x = 0$
d) $||x| - |y|| \leq |x + y| \leq |x| + |y|$

Proposition 5.1.9 (Topology of Ordered Abelian Group)
Let G be an ordered abelian group. Then $U \subset G$ is open in the order topology iff

$$\forall x \in U \exists \varepsilon \in G^+ \text{ s.t. } |y - x| < \varepsilon \implies y \in U$$

Proof. Let $\mathcal{B}$ be the basis defined in (4.1.120). Recall (...) that U is open iff for all $x \in U$ there exists $V \in \mathcal{B}$ such that $x \in V \subseteq U$. If $U \subset G$ satisfies the given condition holds then evidently $x \in \{g \in G \mid y - \varepsilon < g < y + \varepsilon\}$ which is an element of $\mathcal{B}$. Conversely if $x \in V$ with $V = \{g \in G \mid y < g < z\}$, then we may define $\varepsilon = \min(z - x, x - y)$. ■

5.1.1 Sequences in an Ordered Field

Proposition 5.1.10
The order topology on an ordered field K is Hausdorff.

Proposition 5.1.11 (Convergent Sequence)
Let K be an ordered field and (a_n) in K a sequence. Then the following are equivalent

a) $a_n \to \alpha$ *in the sense of (4.1.49)*
b) For all $\varepsilon \in K^+$ there exists $N(\varepsilon) \in \mathbb{N}$ such that $n \geq N(\varepsilon) \implies |a_n - \alpha| < \varepsilon$

As K is Hausdorff limits are unique. When K is Archimedean it is sufficient to consider the case $\varepsilon \coloneqq \frac{1}{N}$ for all positive integers N.

Definition 5.1.12 (Convergent Sequence)
*Let K be an ordered field. We say that a sequence $(a_n)_{n \in \mathbb{N}^+}$ is **cauchy** if for all $\varepsilon > 0$ there exists $N(\varepsilon) > 0$ such that*

$$\forall m, n \in \mathbb{N} : (m, n \geq N(\varepsilon) \implies |a_n - a_m| < \varepsilon)$$

If K is Archimedean then it is sufficient to demonstrate this only for ε of the form $\frac{1}{N}$.

Proposition 5.1.13
Let K be an ordered field. Then every convergent sequence is cauchy.

Proof. Given $\varepsilon > 0$ then $\frac{\varepsilon}{2} > 0$. Therefore there exists N such that

$$n \geq N \implies |a_n - \alpha| < \frac{\varepsilon}{2}$$

Then by the triangle inequality

$$n, m \geq N \implies |a_n - a_m| < \varepsilon$$

■

© David Rufino CC BY-NC-ND 4.0

Definition 5.1.14 (Sequentially Completeness)
*We say an ordered field K is **sequentially complete** if every cauchy sequence is convergent.*

Proposition 5.1.15 (Completeness)
Let K be an ordered field. The following conditions are equivalent

a) Every non-empty subset $X \subset K$ bounded above admits a supremum $\sup X \in K$
b) Every non-empty subset $X \subset K$ bounded below admits an infimum $\inf X \in K$

*In this case we say that K is **complete**.*

Proof. $\inf X = -\sup(-X)$ and vice versa. ∎

Lemma 5.1.16
*Let K be a complete ordered field. Then K is also **Archimedean***

Proof. Suppose K is not Archimedean then $\mathbb{Z}^+$ is bounded above and in particular has a supremum α. Then for every $n \in \mathbb{Z}^+$ we have $n+1 \leq \alpha \implies n \leq \alpha - 1$, which contradicts maximality. ∎

Lemma 5.1.17
Let $X \subset K$ be a subset of an Archimedean ordered field which is "upwards closed" and bounded below (resp. "downwards closed" and bounded above).

Then for every $\varepsilon \in K^+$ there exists an $x \in X$ such that $x - \varepsilon \notin X$ (resp. $x + \varepsilon \notin X$)

Proof. By (5.1.7) there is some $n \in \mathbb{N}^+$ such that $0 < \frac{1}{n} < \varepsilon$. Consider the set

$$\mathcal{S} \coloneqq \left\{ k \in \mathbb{Z} \;\middle|\; \frac{k}{n} \in X \right\}$$

As X is bounded below then so is $\mathcal{S}$. By the well ordering principle it has a minimal element, say k. Define $x \coloneqq \frac{k}{n}$ and then $x - \varepsilon < x - \frac{1}{n} = \frac{k-1}{n}$ which by minimality is not in X, whence neither is $x - \varepsilon$. ∎

Lemma 5.1.18
Let $X \subset K$ be a non-empty subset for which $x \coloneqq \sup X$ exists. Then there exists a sequence (a_n) in X such that $a_n \uparrow x$.

Proof. For every $n \in \mathbb{N}^+$ there exists some $a_n \in X$ such that $a_n > x - \frac{1}{n}$ (otherwise $x - \frac{1}{n}$ would be a smaller upper bound). Then evidently a_n is increasing. For every $\varepsilon \in K^+$ there exists some N such that $0 < \frac{1}{N} < \varepsilon$ by (5.1.7). Consequently

$$n \geq N \implies a_n > x - \frac{1}{N} \implies x \geq a_n > x - \varepsilon \implies |a_n - x| < \varepsilon$$

∎

Definition 5.1.19 (Extended Real Line)
Let K be an ordered field and define the set

$$K^\sharp \coloneqq K \cup \{-\infty\} \cup \{\infty\}$$

with the obvious total ordering and induced order topology. $K \subset K^\sharp$ has the subspace topology and every open subset is of the form

$$U, U \cup (x, \infty], U \cup [-\infty, x), U \cup [-\infty, x) \cup (y, \infty] \; .$$

If $X \subset K$ is a subset which is unbounded above (resp. below) then define $\sup X$ (resp. $\inf X$) to be ∞ (resp. $-\infty$).

Proposition 5.1.20 (Convergence to infinity)
Let K be an ordered field and $a_n \in K^\sharp$ a sequence. Then the following are equivalent

a) $a_n \to \infty$
b) For all $\lambda \in K$ there exists N such that $n \geq N \implies a_n \geq \lambda$

A similar statement holds for $-\infty$.

Lemma 5.1.21 (Limit of Bounded Sequence)
Let $a_n \leq b_n$ be sequences in $K^\sharp$ converging to α and β respectively. Then $\alpha \leq \beta$.

In particular if $a_n \downarrow \alpha$ then $\alpha \leq a_n$.

© David Rufino CC BY-NC-ND 4.0

Proof. If $\beta = \infty$ then the result is vacuously true. If $\beta = -\infty$ then evidently $\alpha = -\infty$ also. Therefore we may assume that β is finite without loss of generality.

Suppose $\alpha > \beta$ and define $\varepsilon = (\alpha - \beta)$. By assumption there exists some N such that

$$n \geq N \implies |a_n - \alpha| < \frac{\varepsilon}{2} \wedge |b_n - \beta| < \frac{\varepsilon}{2}$$

Then $a_n > \frac{\alpha+\beta}{2}$ and $b_n < \frac{\alpha+\beta}{2}$ which is a contradiction. ∎

Proposition 5.1.22 (Subsequence of a Convergent Sequence)
Let K be an ordered field, (a_n) a sequence in $K^\sharp$ and $(n_k)_{k\in\mathbb{N}^+}$ a strictly increasing sequence in $\mathbb{N}^+$. If $a_n \to \alpha$ then $a_{n_k} \to \alpha$ as $k \to \infty$.

Proof. By induction $n_k \geq k$ for all k. Let V be a neighbourhood of α in $K^\sharp$, then by (4.1.49) there exists N such that $n \geq N \implies a_n \in V$. Consequently

$$k \geq N \implies n_k \geq k \geq N \implies a_{n_k} \in V$$

and the result follows. ∎

5.1.2 Sequential Completeness

Lemma 5.1.23 (Monotone Convergence Theorem)
Let K be a complete ordered field and (a_n) an increasing (resp. decreasing) sequence in $K^\sharp$. Then $a_n \to \sup\{a_n\}$. This is finite if and only if a_n is bounded above (resp. below).

Proof. Suppose first that (a_n) is bounded above and define $\alpha := \sup\{a_n\}$. Then for every $\varepsilon \in K^+$ there exists N such that $a_N > \alpha - \varepsilon$ (for otherwise $\alpha - \varepsilon$ would be an upper bound). By assumption $n \geq N \implies \alpha - \varepsilon < a_n \leq \alpha$ whence $|\alpha - a_n| < \varepsilon$. Therefore $a_n \to \alpha$.

The case that (a_n) is unbounded above is trivial. ∎

Proposition 5.1.24 (Complete $\implies$ Sequentially Complete)
*Let K be an ordered field which is **complete**. Then it is **sequentially complete**.*

Proof. Let (a_n) be a cauchy sequence. There exists N such that $n \geq N \implies |a_n - a_N| < 1 \implies |a_n| \leq 1 + |a_N|$. Then $|a_n|$ is bounded by $\max(|a_1|, \cdots, |a_N| + 1)$. Consider the sequence

$$a_n^- := \inf_{k\geq n} a_k$$

Then a_n^- is bounded and increasing. Therefore $a_n^- \uparrow \alpha := \sup a_n^- < \infty$ by (5.1.23). For every $\varepsilon \in K^+$ there exists N_1 such that $n \geq N_1 \implies \alpha - a_n^- < \varepsilon$. There exists N_2 such that $n, m \geq N_2 \implies |a_n - a_m| < \varepsilon$. By definition of a_n^-, for every $n \geq N_1$ there exists $m \geq N_1$ such that $|a_m - a_n^-| < \varepsilon$. Consequently every $n \geq \max(N_1, N_2)$ there is some m such that

$$|\alpha - a_n| \leq |\alpha - a_n^-| + |a_n - a_n^-| \leq |\alpha - a_n^-| + |a_n - a_m| + |a_m - a_n^-| < 3\varepsilon$$

As this is independent of m we see that $a_n \to \alpha$ as required. ∎

Proposition 5.1.25 (Sequentially Complete $\implies$ Complete)
*Let K be an Archimedean ordered field which is **sequentially complete**. Then K is **complete**.*

Proof. Let $X \subset K$ be a non-empty subset bounded above, and $\mathcal{U}$ the set of upper bounds for X, which in particular is bounded below. By assumption $\mathcal{U}$ is non-empty. For each $n \in \mathbb{N}^+$ we may choose $a_n \in \mathcal{U}$ such that $a_n - \frac{1}{2^n} \notin \mathcal{U}$ by (5.1.17). We may assume without loss of generality that a_n is decreasing by setting $a_n := \min_{n'\leq n} a_{n'}$. We argue that

$$a_n - a_{n+1} < \frac{1}{2^n}$$

for otherwise we see that $a_{n+1} \leq a_n - \frac{1}{2^n}$ which implies $a_{n+1} \notin \mathcal{U}$. Therefore by the triangle inequality applied repeatedly

$$m \geq n \implies |a_n - a_m| \leq \sum_{i=n}^{m-1} \frac{1}{2^i} < \frac{1}{2^{n-1}} < \frac{1}{n-1}$$

By the Archimedean property we see that the sequence is cauchy and therefore convergences to $\alpha \in K$. By (5.1.21) $\alpha \leq a_n$. We claim that $\alpha \in \mathcal{U}$. For suppose not then there exists $x \in X$ such that $x > \alpha$. Then there exists n such

© David Rufino CC BY-NC-ND 4.0

that $a_n - \alpha < x - \alpha \implies a_n < x$ which contradicts $a_n \in \mathcal{U}$. Suppose $\exists \alpha' \in \mathcal{U}$ such that $\alpha' < \alpha$. Then for some n we have $\frac{1}{n} < \alpha - \alpha'$ therefore $a_n - \frac{1}{2^n} \geq a_n - \frac{1}{n} \geq \alpha - \frac{1}{n} > \alpha'$. This would imply $a_n - \frac{1}{2^n} \in \mathcal{U}$, a contradiction. Therefore $\alpha' \geq \alpha$ and $\alpha = \sup X$ as required. ∎

5.1.3 Uniqueness of Reals

Lemma 5.1.26
Let K be a complete ordered field. Then every $x \in K$ satisfies

$$x = \sup\left\{\frac{m}{n} \in \mathbb{Q} \;\middle|\; \frac{m}{n} \leq x\right\}$$

Similarly for every $x \in K$ there is a sequence $a_n \in \mathbb{Q}$ such that $a_n \uparrow x$.

Proof. By assumption the supremum α exists. By definition x is an upper bound for the set so $\alpha \leq x$. Suppose $\alpha < x$ then by (5.1.7) there exists $\frac{m}{n}$ such that $\alpha < \frac{m}{n} < x$. On the other hand by definition $\frac{m}{n} \leq \alpha$, which is a contradiction. Therefore $\alpha = x$ as required.

The final statement follows from (5.1.18). ∎

Lemma 5.1.27
Let $\tau : K_1 \hookrightarrow K_2$ be an embedding of ordered fields with K_2 Archimedean. Suppose $x = \sup X$ for some subset $X \subset K_1$. Then $\tau(x) = \sup \tau(X)$.

Proof. Clearly $x' \in X \implies x' \leq x \implies \tau(x') \leq \tau(x) \implies \tau(x) \in \tau(X)^\uparrow$. Suppose $y \in \tau(X)^\uparrow$ and $y < \tau(x)$. Then by (5.1.7) there exists $z \in \mathbb{Q}$ such that $y < z < \tau(x)$. By the order preserving property we have $z \in X^\uparrow$ and $z < x$ which is a contradiction. Therefore $y \in \tau(X)^\uparrow \implies y \geq \tau(x)$ and $\tau(x) = \sup \tau(X)$. ∎

Proposition 5.1.28 (Uniqueness of Reals)
Let K_1, K_2 be complete ordered fields. Then there exists a unique ordered field embedding $K_1 \hookrightarrow K_2$, which is an isomorphism.

Proof. Let $\tau, \tau' : K_1 \to K_2$ be order embeddings. Then they agree on $\mathbb{Q}$. By (5.1.26) every x is the supremum of a set $X \subset \mathbb{Q}$. Then uniqueness follows from (5.1.27). The same result shows it is surjective.

For existence we claim that

$$\tau(x) := \sup\left\{\frac{m}{n} \cdot 1_{K_2} \;\middle|\; \frac{m}{n} \cdot 1_{K_1} \leq x\right\}$$

is a well-defined, order-preserving field isomorphism. Firstly x is bounded by some integer N so the supremum is well-defined. Similarly suppose $x' > x$ then by (5.1.7) we have $x < \frac{m}{n} < x'$. This shows that $\tau(x) \leq \frac{m'}{n'} \leq \tau(x')$ as required, whence τ is order preserving.

By (5.1.26) there exists $a_n, b_n \in \mathbb{Q}$ such that $a_n \uparrow x$, $b_n \uparrow y$ and by the triangle inequality $a_n + b_n \uparrow x + y$. We may show using the definition of τ that $a_n \uparrow \tau(x), b_n \uparrow \tau(y), a_n + b_n \uparrow \tau(x+y)$. Uniqueness of limits shows that $\tau(x+y) = \tau(x) + \tau(y)$. Similarly we may show that $\tau(xy) = \tau(x)\tau(y)$. ∎

5.1.4 Existence of Reals

Proposition 5.1.29 (Construction of the Reals)
There exists a (sequentially) complete ordered field $\mathbb{R}$ which is unique up to a unique order-preserving isomorphism.

*Furthermore it has characteristic zero and the subfield $\mathbb{Q}$ is **order-dense** (5.1.7) and **sequentially dense** (5.1.26).*

*It is **Archimedean** in the sense that for all pairs $x, y \in \mathbb{R}$ there exists $n \in \mathbb{Z}$ such that $nx > y$. Equivalently for all $\varepsilon > 0$ there exists n such that $\frac{1}{n} < \varepsilon$.*

The canonical (order) topology has base the open intervals (a, b), where a, b may be taken to be rational. In particular the order topology on $\mathbb{R}$ is second countable.

Proof. TODO. ∎

5.1.5 n-th Root

For technical reasons it is convenient to introduce the n-th root function at an early stage in an elementary fashion. Other special functions may be defined by similar considerations but it is convenient to wait until there is more machinery available.

© David Rufino CC BY-NC-ND 4.0

Lemma 5.1.30 (Binomial Theorem)
Let K be a field of characteristic 0 and $x, y \in K$. Then

$$(x+y)^n = \sum_{k=0}^{n} \binom{n}{k} x^{n-k} y^k$$

where $\binom{n}{k}$ is a positive integer given by the formula

$$\binom{n}{k} := \frac{n(n-1)\cdots(n-k+1)}{k(k-1)\cdots 1}$$

In particular if K is an ordered field and $x \geq 0$ then for every $0 \leq k \leq n$

$$(1+x)^n \geq \binom{n}{k} x^k$$

Lemma 5.1.31
Let K be a field of characteristic zero, $a, b \in K$ and n a positive integer. Then

$$a^n - b^n = (a-b)(a^{n-1}b + a^{n-2}b^2 + \cdots + ab^{n-2} + b^{n-1})$$

In particular if $a > b \geq 0$ then

$$(a-b)nb^{n-1} < a^n - b^n < (a-b)na^{n-1}$$

Proposition 5.1.32 (n-th root over the Reals)
Let n be a positive integer. Then there exists a function

$$\sqrt[n]{\cdot} : \mathbb{R}_{\geq 0} \to \mathbb{R}_{\geq 0}$$

such that

$$\sqrt[n]{x}^n = \sqrt[n]{x^n} = x$$

It is bijective and strictly increasing.

Proof. Let $a > 0$ be a positive real number. There can be at most one solution because $x > y \implies x^n > y^n$. Let

$$X = \{y \in \mathbb{R}_{\geq 0} \mid y^n \leq a\}$$

then we may see that X is bounded by $\max(1, a)$, for $y > \max(1, a) \implies y^n \geq y > a$.

Define $\alpha = \sup X$. Suppose $\alpha^n > a$ then we claim there is an $h > 0$ such that $(\alpha - h)^n > a$. Then $y > \alpha - h \implies y^n > a$, or contrapositively $y^n \leq a \implies y \leq \alpha - h$, and $\alpha - h$ is an upper bound of X, which is a contradiction. By (5.1.31) we have

$$\alpha^n - (\alpha - h)^n \leq hn\alpha^{n-1}$$

therefore we may choose h such that

$$0 < h < \frac{\alpha^n - a}{n\alpha^{n-1}}\,.$$

Suppose $\alpha^n < a$ then we claim there is $h > 0$ such that $(\alpha + h)^n < a$, which is a contradiction. As before

$$(\alpha + h)^n - \alpha^n \leq hn(\alpha + h)^{n-1}$$

therefore we may choose h such that

$$0 < h < \frac{a - \alpha^n}{n(\alpha + h)^{n-1}}$$

to demonstrate the required property.

By the trichotomy law we deduce that $\alpha^n = a$ as required, and the n-th root function is well-defined and satisfies $\sqrt[n]{x}^n = x$.

Suppose that $a^n = b^n = x > 0$ then evidently $a, b > 0$ and we deduce from (5.1.31) that $a = b$. Therefore the function is unique and injective. It is evidently surjective as $\sqrt[n]{a^n} = a$ by uniqueness. Finally we may deduce from (5.1.31) that for $y > x \geq 0$

© David Rufino CC BY-NC-ND 4.0

$$\sqrt[n]{y} - \sqrt[n]{x} > \frac{y-x}{n\sqrt[n]{y}^{n-1}} > 0$$

and so the function is strictly increasing. ∎

5.1.6 Limsup and Liminf

Proposition 5.1.33 (Limsup and Liminf)
Let K be a complete ordered field and (a_n) a sequence. Define the associated sequences in $K^\sharp$

$$a_n^+ := \sup_{n' \geq n} a_{n'}$$
$$a_n^- := \inf_{n' \geq n} a_{n'}$$

which are decreasing and increasing sequences respectively. Define

$$\limsup a_n = \lim_{n\to\infty} a_n^+$$
$$\liminf a_n := \lim_{n\to\infty} a_n^-$$

Then

a) a_n is bounded above $\iff a_n^+$ is bounded above $\iff \limsup a_n^+ < \infty$
b) a_n is bounded below $\iff a_n^-$ is bounded above $\iff \liminf a_n^- > -\infty$
c) $\limsup a_n \geq \liminf a_n$
d) $a_n \to \alpha \iff \limsup a_n = \liminf a_n = \alpha$

Proof. Clearly a_n^+ is decreasing and a_n^- is increasing. Therefore by (5.1.23) the definition of $\limsup a_n$ and $\liminf a_n$ is well-defined, the same result also demonstrates a) and b). By definition $a_n^+ \geq a_n \geq a_n^-$ so the inequality c) follows from (5.1.21).

Suppose $\alpha \to \infty$ then for every $L > 0$ there exists N such that

$$n \geq N \implies a_n > L \implies a_n^- \geq L$$

whence $\liminf a_n = \infty$. Conversely suppose $\liminf a_n = \infty$ we see by definition that $a_n \to \infty$ as required. The case of $-\infty$ follows similarly.

Suppose that $\alpha = \limsup a_n = \liminf a_n$ is finite, then there exists N_1 such that

$$n \geq N_1 \implies a_n \geq a_n^- > \alpha - \frac{\varepsilon}{2}$$

and N_2 such that

$$n \geq N_2 \implies a_n \leq a_n^+ < \alpha + \frac{\varepsilon}{2}$$

Then

$$n \geq \max(N_1, N_2) \implies |a_n - \alpha| < \varepsilon$$

and we conclude $\lim_{n\to\infty} a_n = \alpha$.

Conversely suppose $a_n \to \alpha$. Then for every $\varepsilon > 0$ there exists some N such that

$$n \geq N \implies \alpha - \varepsilon < a_n < \alpha + \varepsilon$$

and therefore

$$n \geq N \implies \alpha - \varepsilon \leq a_n^- \leq a_n \leq a_n^+ \leq \alpha + \varepsilon$$

Evidently then

$$\alpha - \varepsilon \leq \liminf a_n \leq \limsup a_n \leq \alpha + \varepsilon$$

As ε was arbitrary then this shows $\alpha = \liminf a_n = \limsup a_n$. ∎

Proposition 5.1.34 (Arithmetic Properties of limsup and liminf)
Let K be a complete ordered field and $(a_n), (b_n)$ bounded sequences. Then

a) $\limsup(a_n + b_n) \leq \limsup a_n + \limsup b_n$

© David Rufino CC BY-NC-ND 4.0

b) $\liminf(a_n + b_n) \leq \liminf a_n + \liminf b_n$

If $b_n \to b$ *then*

c) $\limsup(a_n + b_n) = \limsup a_n + b$
d) $\liminf(a_n + b_n) = \liminf a_n + b$

and if in addition $b \geq 0$ *then*

e) $\limsup(a_n b_n) = b \limsup a_n$
f) $\liminf(a_n b_n) = b \liminf a_n$

Proof. We prove each in turn

a) By definition $n' \geq n \implies a_{n'} + b_{n'} \leq a_n^+ + b_n^+$, whence $(a_n + b_n)^+ \leq a^+ + b_n^+$. The result follows from (5.1.21).
b) Similarly
c) By a) we have $\limsup(a_n + b_n) \leq \limsup a_n + b$. Furthermore $\limsup(a_n) = \limsup(a_n + b_n + (-b_n)) \leq \limsup(a_n + b_n) + \limsup(-b_n) = \limsup(a_n + b_n) - b$, and the result follows.
d) Similarly

e), f) Suppose first that $b = 0$. Then for sufficiently large n

$$0 \leq b_n \leq \varepsilon \implies -\varepsilon|a_n^-| \leq (a_n b_n)^+ \leq |a_n^+|\varepsilon \implies \left|(a_n b_n)^+\right| \leq \max\left(\left|\sup_k a_k^+\right|, \left|\inf_k a_k^-\right|\right)\varepsilon$$

as (a_n) is assumed bounded then this shows that $(a_n b_n)^+ \to 0$ as required. In this case f) follows similarly. For the general case then

$$a_n b_n = a_n(b_n - b) + a_n b$$

Obviously $b_n - b \to 0$, so by the case already proven we have

$$\limsup\left(a_{n(b_n - b)}\right) = \liminf\left(a_{n(b_n - b)}\right) = 0$$

and in particular $a_n(b_n - b) \to 0$ by (5.1.33).d). Consequently by c)

$$\limsup(a_n b_n) = \limsup(a_n b) = b \limsup(a_n)$$

as required. The case f) follows similarly. ∎

Proposition 5.1.35
Let K *be a complete ordered field and* (a_n) *a bounded sequence. Then every subsequence* $\left(a_{n_k}\right)$ *satisfies*

$$\limsup_k a_{n_k} \leq \limsup_n a_n$$

and there in fact exists a convergent subsequence a_{n_k} *such that*

$$a_{n_k} \downarrow \limsup_n a_n$$

A similar statement applies to $\liminf a_n$.

Proof. Define $b_k \coloneqq \sup_{k' \geq k} a_{n_{k'}}$, then by definition

$$b_k^+ \leq a_{n_k}^+$$

Taking limits as $k \to \infty$ and considering we find that

$$\limsup_k a_{n_k} \overset{(5.1.21)}{\leq} \lim_{k \to \infty} a_{n_k}^+ \overset{(5.1.22)}{=} \lim_{n \to \infty} a_n^+ =: \limsup_n a_n$$

Recall that $a_n^+ \downarrow \limsup a_n$. Therefore for every $k > 0$ there exists N_k such that $a_{N_k}^+ \leq \limsup a_n + \frac{1}{2k}$. Then by definition there exists some $n_k \geq N_k$ such that $a_{n_k} \leq a_{N_k}^+ + \frac{1}{2k} \leq \limsup a_n + \frac{1}{k}$. We see that a_{n_k} has the required property. ∎

© David Rufino CC BY-NC-ND 4.0